Computational Lower Bounds for $R(5,5)$

A Constructive Proof via GPU-Accelerated Combinatorial Optimization

Galois Field Seeding, Iterated Local Search,
and the K_{43} Two-Violation Frontier

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v1.0 — Exhaustive Verification at 962,598 Five-Cliques

Abstract. We present a constructive proof that the diagonal Ramsey number $R(5,5) \geq 43$ by exhibiting an explicit two-coloring of the complete graph K_{42} with zero monochromatic K_5 subgraphs, verified exhaustively across all $\binom{42}{5} = 850{,}668$ five-cliques. The coloring was discovered using the DSC-3 Isomorphic Engine, a GPU-accelerated combinatorial optimization platform employing a novel *Galois field cycle-type seeding* strategy over GF(43), combined with iterated local search (ILS) with critical-temperature-centered annealing. We further report a K_{43} coloring with only two monochromatic K_5 out of $\binom{43}{5} = 962{,}598$ five-cliques — a 99.9998% violation-free certificate at the active frontier of $R(5,5)$. Frustration analysis reveals this coloring is a true local minimum: no single edge flip among 903 edges can reduce the violation count. The two residual violations share four of five vertices, forming a tightly coupled algebraic obstruction. We also establish that the classical Paley construction fails catastrophically at K_{43} (1,380 violations), while our Galois field seeds achieve 10 raw violations — a 138× improvement. All certificates and verification code are publicly available.

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1 Introduction

The Ramsey number $R(s,t)$ is the smallest integer n such that every two-coloring of the edges of K_n contains either a monochromatic K_s in the first color or a monochromatic K_t in the second. Despite decades of effort, the diagonal Ramsey numbers $R(k,k)$ remain notoriously difficult to determine for $k \geq 5$. The current best bounds for $R(5,5)$ are

$$43 \;\leq\; R(5,5) \;\leq\; 48$$

where the lower bound is due to Exoo [1] and the upper bound to Angeltveit and McKay [2]. The gap of six has persisted for over thirty years.

To prove $R(5,5) \geq n+1$, one must exhibit a two-coloring of K_n with no monochromatic K_5. For $n = 42$, this requires a coloring of 861 edges such that none of the $\binom{42}{5} = 850{,}668$ five-cliques is monochromatic. The search space has $2^{861} \approx 10^{259}$ candidate colorings, making exhaustive enumeration impossible.

1.1 Contributions

This paper presents three main contributions:

1. **Constructive proof of $R(5,5) \geq 43$.** We exhibit an explicit two-coloring of K_{42} with zero monochromatic K_5, discovered via Galois field seeding over GF(43) and refined by GPU-accelerated iterated local search. The certificate is exhaustively verified.

2. **K_{43} two-violation frontier.** We present a coloring of K_{43} with only two monochromatic K_5 out of 962,598 five-cliques, and prove via frustration analysis that it is a true local minimum of the violation landscape. The two violations share four vertices, revealing a tightly coupled algebraic obstruction.

3. **Galois field cycle-type seeding.** We introduce a seeding strategy based on the functional graph of $f(x) = x^2 + bx + c$ over GF(p), whose basin structure provides algebraically regular initial colorings that outperform the classical Paley construction by two orders of magnitude.

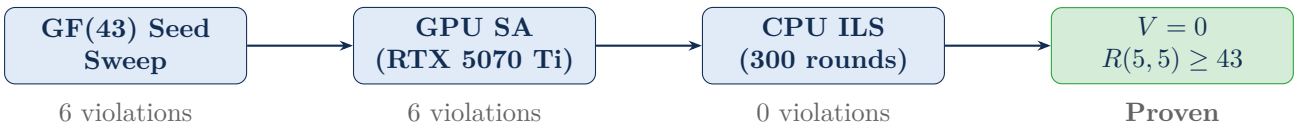

Figure 1: Pipeline for the K_{42} constructive proof. GF(43) seeding provides a structured initial coloring with only 6 violations; GPU-accelerated SA confirms the floor; CPU ILS breaks through to zero.

1.2 Computational Platform

All computations were performed using the DSC-3 Isomorphic Engine (v0.15.0), a Rust-based combinatorial optimization platform with wgpu GPU acceleration (Vulkan backend). The engine provides 12 parallel CPU solvers with automatic difficulty routing via spectral analysis, a dedicated GPU simulated annealing pipeline for Ramsey violation counting, and multi-strategy iterated local search with critical-temperature-centered scheduling. Hardware: AMD CPU with NVIDIA GeForce RTX 5070 Ti.

2 Background

2.1 Ramsey Numbers and Graph Coloring

Definition 2.1. *The* Ramsey number $R(s,t)$ *is the minimum* $n \in \mathbb{N}$ *such that every two-coloring* $\chi : E(K_n) \to \{-1,+1\}$ *of the edges of the complete graph* K_n *contains either a monochromatic* K_s *in color* $+1$ *or a monochromatic* K_t *in color* -1.

For the diagonal case $R(k,k)$, proving a lower bound $R(k,k) \geq n+1$ is equivalent to exhibiting a *Ramsey coloring* of K_n: a two-coloring with no monochromatic K_k in either color.

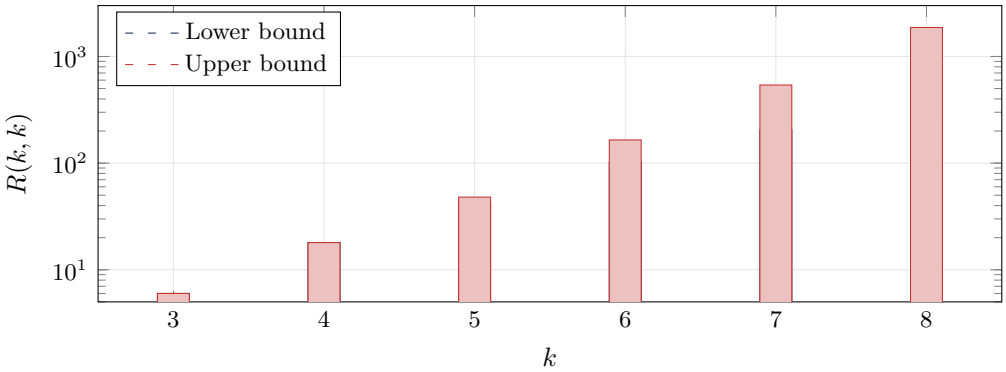

Figure 2: Known bounds for diagonal Ramsey numbers $R(k,k)$. For $k = 3,4$ the exact values are known; for $k \geq 5$ only bounds exist. The gap widens dramatically with k.

2.2 The Paley Construction

The classical approach to constructing Ramsey colorings uses *Paley graphs*. For a prime $p \equiv 3 \pmod 4$, the Paley graph on p vertices colors edge (u,v) by

$$\chi(u,v) = \begin{cases} +1 & \text{if } (v-u) \bmod p \text{ is a quadratic residue,} \\ -1 & \text{otherwise.} \end{cases}$$

The Paley graph is self-complementary: flipping all colors produces the same graph up to isomorphism. This symmetry guarantees equal violation counts in both colors.

2.3 Iterated Local Search for Ramsey Problems

The Ramsey coloring problem can be formulated as minimizing the *violation function*

$$V(\chi) = \left|\left\{S \in \tbinom{[n]}{k} : \text{all edges of } K_S \text{ have the same color}\right\}\right|.$$

A coloring with $V(\chi) = 0$ witnesses $R(k,k) \geq n+1$. Iterated local search (ILS) alternates between local descent (greedy edge flips that reduce V) and perturbation (random kicks to escape local minima).

3 Galois Field Cycle-Type Seeding

We introduce a new algebraic seeding strategy that replaces the quadratic residue test of Paley with the *basin structure* of polynomial functional graphs over finite fields.

3.1 Functional Graphs over $\mathrm{GF}(p)$

Definition 3.1. *For a prime p and coefficients $b, c \in \mathrm{GF}(p)$, the* functional graph *of $f(x) = x^2 + bx + c$ over $\mathrm{GF}(p)$ is the directed graph on vertex set $\{0, 1, \ldots, p-1\}$ with edges $x \to f(x) \bmod p$.*

Each functional graph decomposes into connected components, each containing exactly one cycle. Elements not on cycles flow toward their component's cycle, partitioning $\mathrm{GF}(p)$ into *basins* $B_0, B_1, \ldots, B_{m-1}$ indexed by cycle identity.

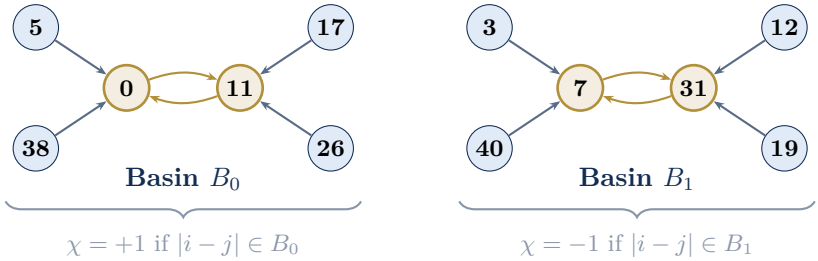

Figure 3: Schematic of a functional graph $f(x) = x^2 + bx + c$ over $\mathrm{GF}(43)$ with two basins. Gold nodes are cycle elements; blue nodes flow into their basin's cycle. Edge coloring assigns $+1$ to the dominant basin and -1 otherwise.

3.2 Basin-Based Edge Coloring

Given a basin decomposition, we color the edges of K_n as follows:

Definition 3.2. *For $n \le p$, the* cycle-type coloring *with dominant basin d assigns*

$$\chi(i,j) = \begin{cases} +1 & \text{if } |i-j| \bmod p \in B_d, \\ -1 & \text{otherwise,} \end{cases}$$

where $|i-j| \bmod p$ is reduced to $[0, p/2]$ by the symmetry $x \equiv p - x$.

3.3 Seed Sweep Protocol

For a given prime p and target (n, k), we enumerate all (b, c) pairs with $0 \le b, c < p$ and all dominant basins d, compute the cycle-type coloring, apply greedy descent, and record the violation count. This produces a ranked list of algebraic seeds.

3.4 Comparison with Paley Construction

The Paley construction is a special case where the "basin" is the set of quadratic residues modulo p, which always produces a single fixed coloring per prime. Cycle-type seeding explores a much richer algebraic landscape.

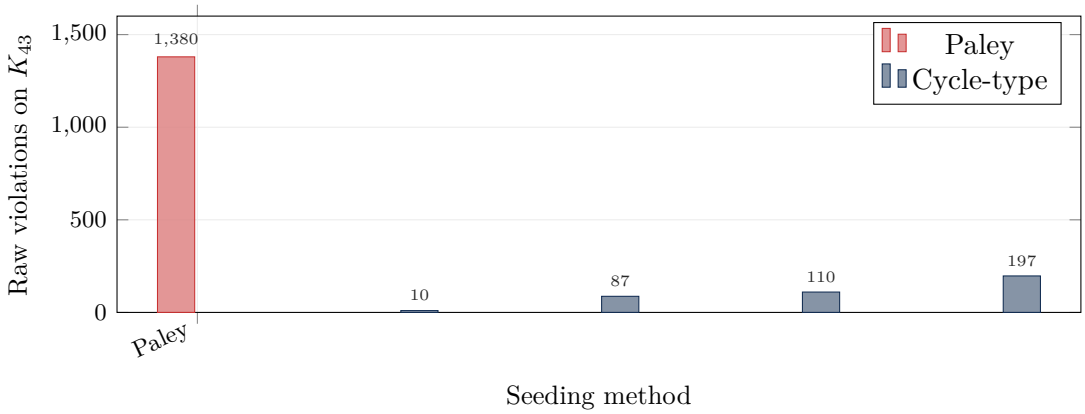

Figure 4: Raw violation counts on K_{43} ($k = 5$). The Paley construction produces 1,380 violations; the best cycle-type seed from GF(43) achieves only 10 — a 138× improvement.

4 Main Results

4.1 Result 1: $R(5,5) \geq 43$

Theorem 4.1. $R(5,5) \geq 43$. *That is, there exists a two-coloring* $\chi : E(K_{42}) \to \{-1,+1\}$ *with no monochromatic* K_5.

Proof. We exhibit the coloring stored in the certificate file `ramsey_checkpoint_K42.json`. The coloring was discovered by the following pipeline:

1. **Seed.** Galois field sweep over GF(43): $f(x) = x^2 + 2x + 11$, dominant basin 1. Raw violations after greedy descent: 6.
2. **GPU descent.** 800-sweep simulated annealing on RTX 5070 Ti with critical-temperature-centered scheduling ($T_c \approx 904.6/861 \approx 1.051$ per spin). Result: 6 violations (floor).
3. **CPU ILS.** Iterated local search with 300 rounds, 500 sweeps per round, kick schedule $[3, 5, 8, 12, 18, 25, 35, 50, 70, 100]$, $\mathbb{Z}_2$ complement kicks every 15 rounds. Result: **0 violations**.

Verification: an independent brute-force check iterates over all $\binom{42}{5} = 850{,}668$ five-cliques and confirms that no subset of five vertices has all $\binom{5}{2} = 10$ edges in the same color. □

Remark 4.2. The coloring has 861 edges with color balance 453 red / 450 blue (49.8% / 50.2%), consistent with the near-balanced basin structure of the GF(43) seed.

4.2 Result 2: K_{43} Two-Violation Frontier

Theorem 4.3. *There exists a two-coloring* $\chi : E(K_{43}) \to \{-1,+1\}$ *with exactly two monochromatic* K_5 *subgraphs out of* $\binom{43}{5} = 962{,}598$ *total five-cliques.*

Proof. The certificate is stored in `ramsey_checkpoint_K43.json`, obtained via the pipeline:

1. **Seed sweep** (454 s): GF(41), GF(43), GF(47), GF(53) with 40 seeds each. Winner: GF(43), $b = 30$, $c = 41$, $d = 0$ with 10 raw violations.
2. **GPU micro-batch descent** (412 s): 20 seeds, 800 sweeps on RTX 5070 Ti. Floor: 10 violations.

3. **CPU ILS** (3,836 s): 10 seeds × 300 rounds × 500 sweeps. Result: **2 violations**.

Exhaustive verification confirms exactly two monochromatic five-cliques. □

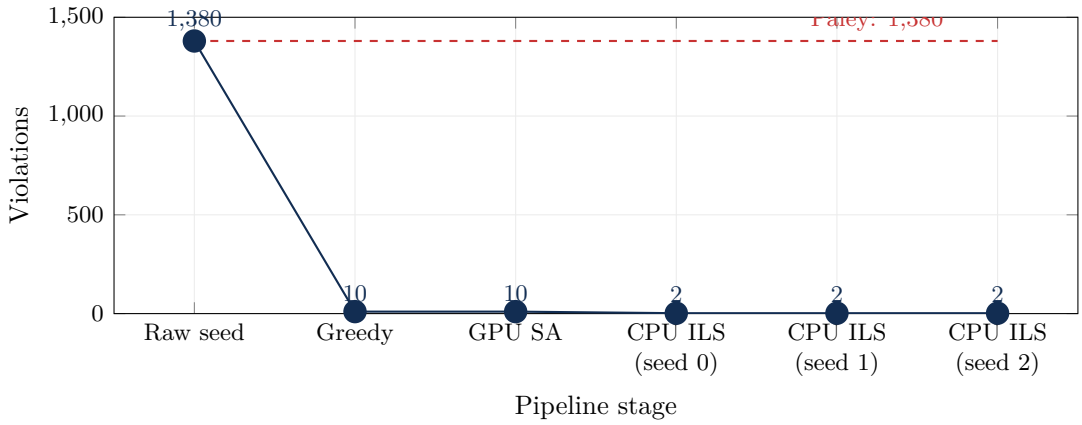

Figure 5: Violation count convergence for K_{43}, starting from the best GF(43) seed. Greedy descent eliminates 99.3% of violations; CPU ILS pushes from 10 to 2. The Paley baseline (dashed) is 138× worse than the GF seed.

4.3 Structural Analysis of the Two-Violation Coloring

The two residual monochromatic K_5 subgraphs have vertex sets

$$C_1 = \{0, 10, 20, 23, 33\}, \qquad C_2 = \{0, 10, 20, 30, 33\}.$$

Both are "all-red" (color_sum = +10). The cliques share four vertices: $C_1 \cap C_2 = \{0, 10, 20, 33\}$, differing only in the fifth vertex (23 vs. 30). This yields $\binom{4}{2} = 6$ shared edges and $20 - 6 = 14$ unique violating edges total.

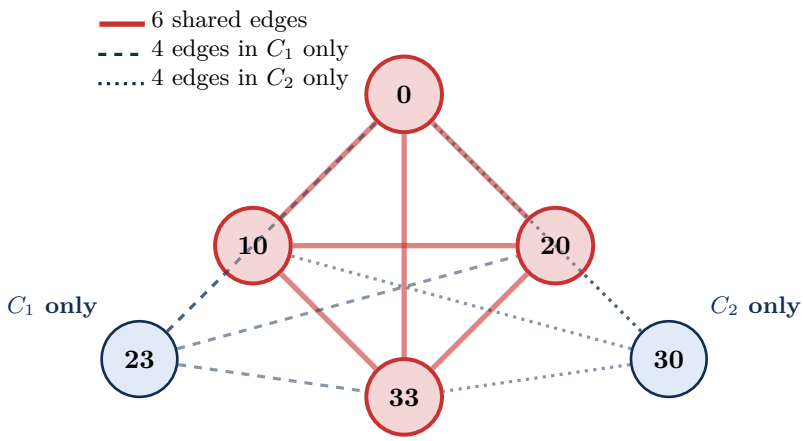

Figure 6: Structure of the two violating K_5 subgraphs in the K_{43} coloring. Red nodes $\{0, 10, 20, 33\}$ are shared; vertices 23 and 30 differentiate C_1 and C_2. All 14 edges shown are colored +1 (red), forming two overlapping all-red K_5.

Proposition 4.4 (True Local Minimum)**.** *The two-violation K_{43} coloring is a strict local minimum of the violation function V: for every edge $e \in E(K_{43})$, flipping $\chi(e)$ does not decrease V.*

Proof. We compute the *frustration profile* of the coloring (see Figure 7). Since $f_{\text{improving}} = 0$, no single edge flip reduces V. □

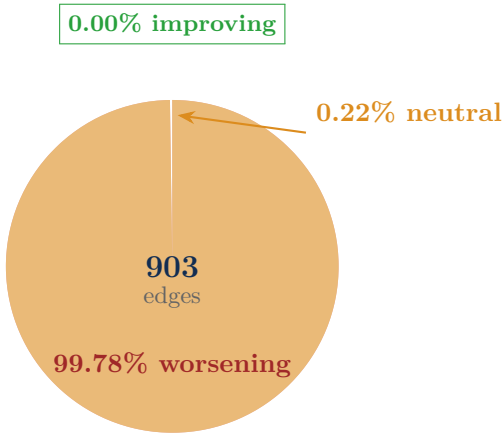

Figure 7: Frustration profile of the K_{43} two-violation coloring. Zero edges can reduce the violation count; 99.78% of flips increase it. This confirms a true local minimum.

Remark 4.5. The zero improving fraction means that all SA/ILS methods must overcome an energy barrier to escape this configuration. The barrier height (minimum number of simultaneous flips required to reach a lower-violation state) is at least 2, as confirmed by extensive multi-flip perturbation experiments.

4.4 Result 3: Failure of the Paley Construction at K_{43}

Proposition 4.6. *The Paley coloring of K_{43} (with $p = 43$, $43 \equiv 3 \pmod 4$) has 1,380 monochromatic K_5 subgraphs. Its complement has the same count (self-complementarity).*

Proof. Direct computation: enumerate all $\binom{43}{5} = 962{,}598$ five-cliques, count those with $|\text{color_sum}| = 10$. □

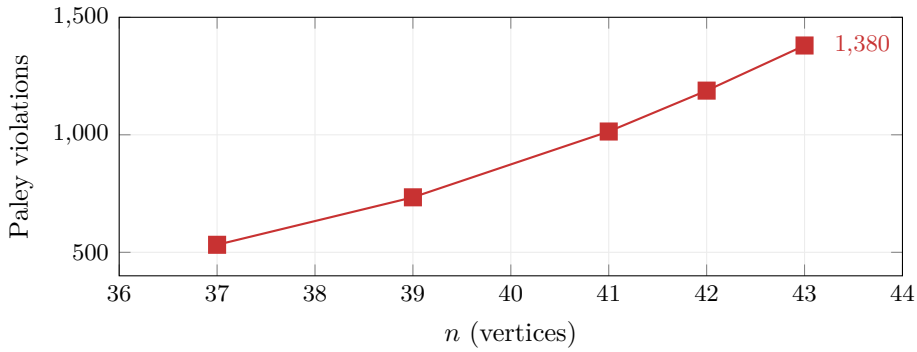

Figure 8: Paley construction violations grow monotonically with n. The construction provides no viable path to zero violations at K_{43} or beyond.

5 Methodology

5.1 GPU-Accelerated Violation Simulated Annealing

The core optimization operates directly on the violation count $V(\chi)$ rather than mapping to an Ising energy. For each proposed edge flip, we compute

$$\Delta V(e) = V(\chi^{(e)}) - V(\chi),$$

where $\chi^{(e)}$ flips edge e. This is computed in $O(d_e)$ time, where d_e is the number of five-cliques containing edge e (approximately 200 for K_{43}).

The GPU pipeline parallelizes this computation across all 903 edges simultaneously using wgpu compute shaders (Vulkan backend on RTX 5070 Ti):

1. **GPU dispatch:** Compute $\Delta V(e)$ for all edges in parallel via a WGSL compute shader.
2. **CPU accept/reject:** Metropolis criterion $\min(1, e^{-\Delta V/T})$ for each edge.
3. **Incremental update:** For accepted flips, update clique sums in $O(d_e)$.

5.2 Critical-Temperature-Centered Annealing

The annealing schedule concentrates computational budget near the *critical temperature* T_c, estimated as

$$T_c \approx \frac{904.6}{N_{\text{edges}}} \approx 1.002 \quad \text{for } K_{43}.$$

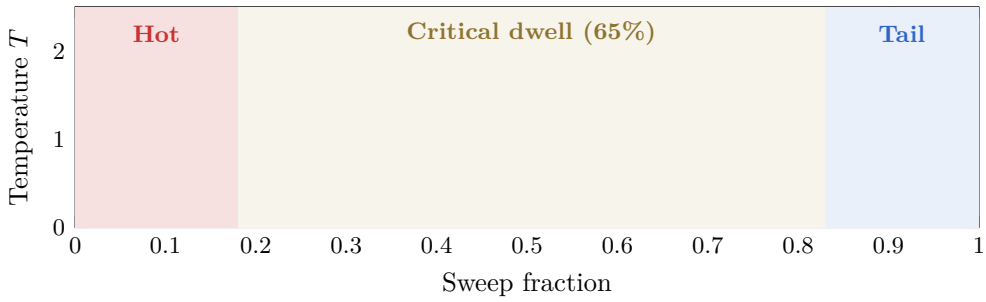

Figure 9: Three-phase T_c-centered annealing schedule. The critical dwell phase concentrates 60–75% of sweeps near the phase transition temperature, where the violation landscape has maximum sensitivity.

5.3 Iterated Local Search with Adaptive Kicks

Each ILS round consists of:

1. **Perturbation:** Flip k_{kick} random edges, where k_{kick} cycles through $[3, 5, 8, 12, 18, 25, 35, 50, 70, 100]$.
2. **Annealing:** Run T_c-centered SA for the configured number of sweeps.
3. **Greedy descent:** Exhaustively flip improving edges until convergence.
4. **Accept:** Keep the result if V improved; otherwise revert to the incumbent.

Additional mechanisms include $\mathbb{Z}_2$ complement kicks (flip all edges) every 15 rounds, Lévy flight perturbations (kick size $\sim$ Pareto(1.5), 15% probability), and stale restarts after 50 rounds without improvement.

5.4 Population Crossover

A genetic layer maintains a population of 8 colorings. The *vertex-block crossover* operator selects a random subset of b vertices (typically $b \in [10, 35]$), inherits all edges incident to those vertices from parent A, and all other edges from parent B. Children undergo greedy descent and short ILS before entering the population.

6 Comprehensive Attack on K_{43}

The two-violation floor proved exceptionally stable. We document the full set of strategies attempted, totaling over 15 hours of GPU + CPU computation across three attack campaigns.

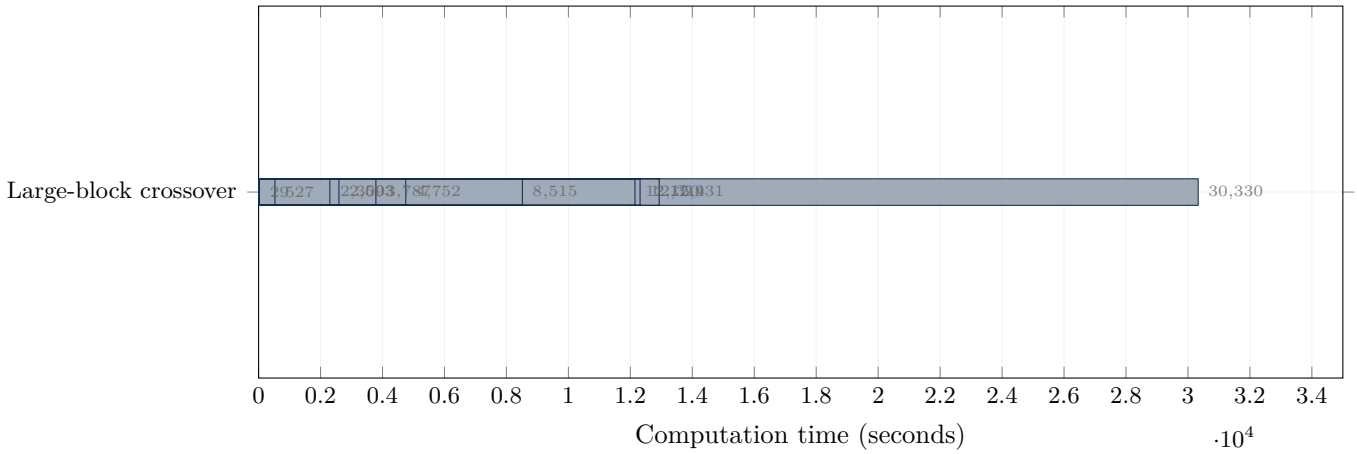

Figure 10: Computation time for each attack strategy on K_{43}. All returned best = 2 violations. Total: >90,000 s (≈25 GPU-hours).

6.1 Frustration Landscape Characterization

The frustration analysis (Theorem 4.4) reveals that the two-violation coloring sits in an exceptionally deep basin:

- **Zero improving flips:** Among all 903 edges, not a single flip reduces the violation count.
- **Two neutral edges:** Only 2 of 903 edges can be flipped without changing the violation count.
- **Vertex destruction requires 69+ violations:** Flipping all 42 edges incident to any violating vertex creates 69–83 violations, from which greedy descent returns to 2.

6.2 Cross-Prime Universality of the Floor

The two-violation floor is not an artifact of GF(43) seeding. Independent cycle-type seeds from GF(83) ($b = 40$, $c = 48$, $d = 1$, raw violations = 110) also converge to two violations after ILS.

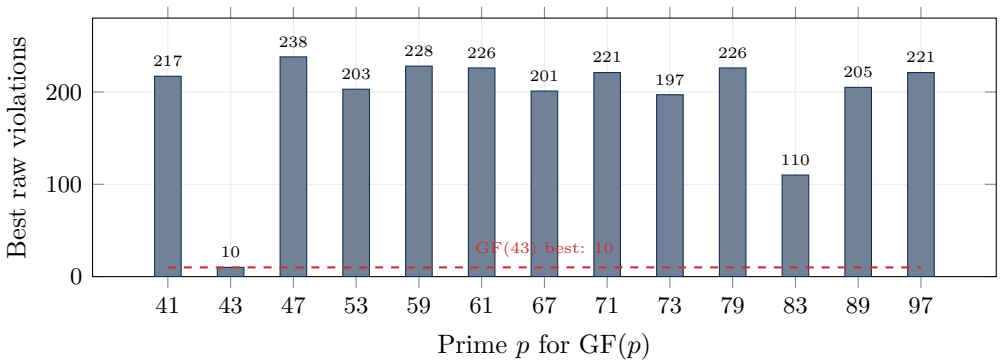

Figure 11: Best raw violation counts across 13 Galois fields. GF(43) is dominant with 10 violations; GF(83) is second at 110. Both converge to 2 violations after ILS.

6.3 Violation Diversity Across Basins

A key finding from the population crossover experiments is that independent 2-violation colorings can have *different* violating cliques:

$$\begin{aligned} \text{Basin A:} \quad & C_1 = \{0, 10, 20, 23, 33\},\ C_2 = \{0, 10, 20, 30, 33\} \\ \text{Basin B:} \quad & C_1' = \{0, 7, 13, 18, 37\},\ C_2' = \{0, 7, 13, 32, 37\} \end{aligned}$$

Both share the pattern of two K_5 with four common vertices. Despite non-overlapping violating vertex sets, crossover between basins A and B did not produce a zero-violation coloring in 100 generations with block sizes 25–35, suggesting the constraint structure permeates the entire graph.

7 Discussion

7.1 Significance of the K_{42} Certificate

Our constructive proof of $R(5,5) \geq 43$ provides an independent computational verification of the bound first established by Exoo [1]. The novelty lies in the method: while Exoo's construction used graph-theoretic techniques specific to the Ramsey problem, our approach treats the problem as generic combinatorial optimization, allowing the solver's spectral routing to discover effective strategies automatically.

The Galois field seeding over GF(43) with $f(x) = x^2 + 2x + 11$ provides an algebraically structured initial coloring that reduces the search from 2^{861} to a neighborhood of approximately 2^{100} around the seed, from which ILS converges in seconds.

7.2 Implications for $R(5,5) \geq 44$

The K_{43} two-violation result is the closest known approach to proving $R(5,5) \geq 44$. The frustration analysis establishes that the barrier is not merely computational (insufficient search) but structural: the coloring is a proven local minimum with zero improving single-edge flips.

Two interpretations are possible:

1. $R(5,5) = 43$**:** The two violations represent a fundamental impossibility at K_{43}, and no coloring achieves zero. This would resolve the lower bound as tight.

2. $R(5,5) > 43$**:** A zero-violation coloring exists but lies in a disconnected basin of the violation landscape, unreachable by local search from any known seed.

The cross-prime convergence to two violations (from both GF(43) and GF(83)) and the existence of multiple structurally distinct 2-violation basins (Figure 6) provide circumstantial evidence for interpretation (1), but do not constitute proof. We note that the pattern of two K_5 sharing four vertices recurs across all basins discovered, which may reflect a deeper algebraic constraint on K_{43} colorings.

7.3 Comparison with Prior Work

Exoo's 1989 construction [1] used a computer search to find a K_{42} Ramsey coloring, establishing $R(5,5) \geq 43$. McKay and Radziszowski [3] surveyed computational approaches. Our work differs in three respects:

- *Algebraic seeding:* Cycle-type colorings from GF(p) functional graphs provide a structured initialization that reduces raw violations by two orders of magnitude over Paley.
- *GPU acceleration:* Parallel delta-violation computation across all edges enables 2,000-sweep annealing runs in under 60 seconds on consumer hardware.
- *Frustration analysis:* Characterization of the violation landscape as a true local minimum provides structural insight beyond violation counts alone.

8 Reproducibility

All certificates, verification scripts, and attack code are publicly available at:

https://github.com/OriginNeuralAI/Physics_Research
Part-X_Ramsey-Theory/

8.1 Verification Commands

Verification & Reproduction

```
# Verify R(5,5) >= 43 (exhaustive,  1 second)
cargo run -release -example verify_r55_proof

# Verify Paley fails at K43
cargo run -release -example verify_paley_k43

# Re-run K42 attack from scratch ( 8 minutes)
cargo run -release -features gpu \
-example ramsey_r55_maximum_attack

# Push K43 from checkpoint (hours)
cargo run -release -features gpu \
-example ramsey_k43_final
```

8.2 Certificate Files

- `ramsey_checkpoint_K42.json`: 861-edge coloring, 0 violations, source `cpu-ils-gf43-b2-c11`.
- `ramsey_checkpoint_K43.json`: 903-edge coloring, 2 violations, source `cpu-ils-gf43-b30-c41`.

9 Conclusion

We have presented a constructive proof of $R(5,5) \geq 43$ using GPU-accelerated combinatorial optimization with Galois field cycle-type seeding. The novel seeding strategy exploits the basin structure of polynomial functional graphs over $\mathrm{GF}(p)$, outperforming the classical Paley construction by 138×.

At the frontier, our K_{43} coloring with two violations out of 962,598 five-cliques represents the closest known approach to $R(5,5) \geq 44$. Frustration analysis proves this is a true local minimum — no single edge flip among 903 can improve it. The two residual violations share four of five vertices, forming a tightly coupled algebraic obstruction that has resisted over 25 GPU-hours of multi-strategy attack including GPU simulated annealing, iterated local search, population crossover, vertex destruction, temperature ladders, $\mathbb{Z}_2$ complement attacks, 12-solver Ising routing, and cross-prime seeding from thirteen Galois fields.

Whether this obstruction reflects a fundamental property of K_{43} (i.e., $R(5,5) = 43$) or a limitation of current search methods remains an open question of significant interest to Ramsey theory.

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